

CNES Cloud Chamber

Control PCB

PCB Design Report

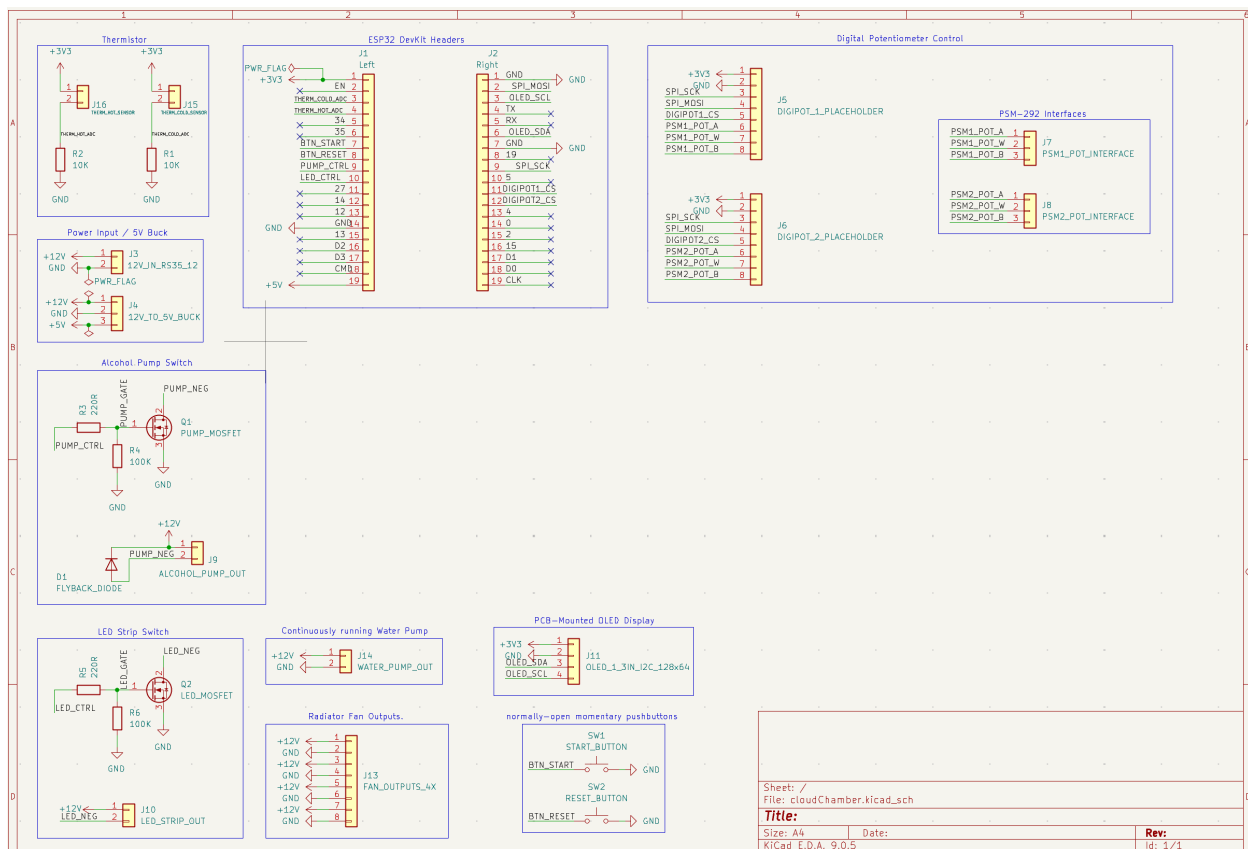
Concordia Nuclear Engineering Society

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Design snapshot - 12 September 2026

Revision 0.2

An ESP32 carrier board for temperature sensing, auxiliary load control and the planned electronic adjustment of Peltier power.



Current control PCB schematic

Development stage: schematic refinement before footprint assignment and PCB layout.

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Project objective and current status

The CNES portable diffusion cloud chamber is intended to make ionizing particle tracks visible in supersaturated alcohol vapor above a cooled plate. This work focuses on the development of the low-voltage control PCB that connects temperature sensors, auxiliary loads and the user interface to an ESP32 DevKit.

The current design establishes the carrier-board architecture, pin assignments, thermistor dividers, pump and LED switches, and continuous fan and water-pump outputs. Schematic cleanup reached a reported ERC result of zero active violations. Digital potentiometer compatibility, final component selection, footprint assignment and PCB routing remain to be completed. This report records the design in progress for CNES project continuity and future reference. PCB manufacture and hardware validation remain pending.

The PCB reads hot-side and cold-side temperatures, switches the alcohol pump and bezel lighting, supports an OLED and two buttons, and distributes auxiliary 12 V power. Camera trail classification, permanent count storage and the high voltage clearing field remain broader system functions without finalized implementations on this board. The eventual display requirement includes plate temperature, timer, alpha and beta totals, and counts per minute for each; the initial display focus is cold-plate temperature.

The mains and high-current Peltier paths remain external. The PCB carries auxiliary load current and control signals; it does not route the 24 V Peltier power.

System architecture

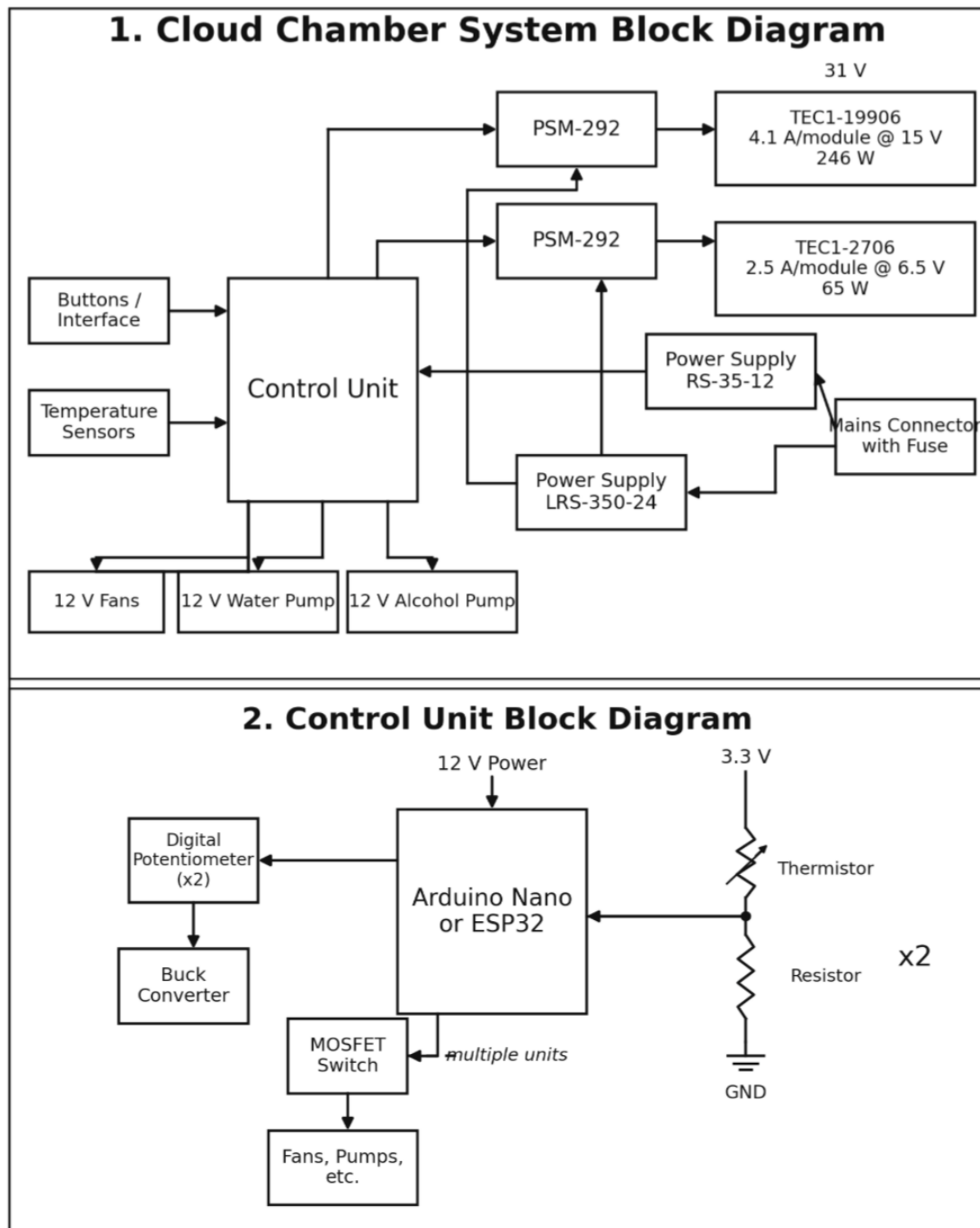


Figure 1. Overall system block diagram.

The high-level power architecture discussed is:

The fused and switched mains input feeds two separate supplies: the Mean Well RS-35-12 supplies the 12 V auxiliary branch, and the Mean Well LRS-350-24 supplies the 24 V Peltier branch.

The 12 V path powers the control PCB, pumps, fans, LED strip, and a 12 V to 5 V buck stage.

The 24 V path powers the PSM-292 buck converters and Peltier modules. The 24 V / Peltier current is not routed through this PCB.

Known system-level parts from the project/BOM context:

- TEC1-12706 Peltier modules, quantity 4, listed as first-stage Peltier modules
- TEC1-19906 Peltier modules, quantity 4, listed as second-stage Peltier modules
- Mean Well LRS-350-24 power supply, quantity 1, for 24 V Peltier power
- Mean Well RS-35-12 power supply, quantity 1, for 12 V control/auxiliary electronics
- PSM-292 adjustable buck converter modules, quantity 2
- Alcohol pump, 12 V / 5 W, final according to Nathaniel
- Water pump, closer to 20 W according to Nathaniel
- Radiator fans, 120 mm, exact part still not finalized in BOM
- LED strip for bezel, still listed as TBD
- Generic thermistors, quantity 2
- OLED screen and pushbuttons, to be mounted on PCB
- 12 V to 5 V buck converter, exact module/implementation still unresolved

ESP32 DevKit Header Design

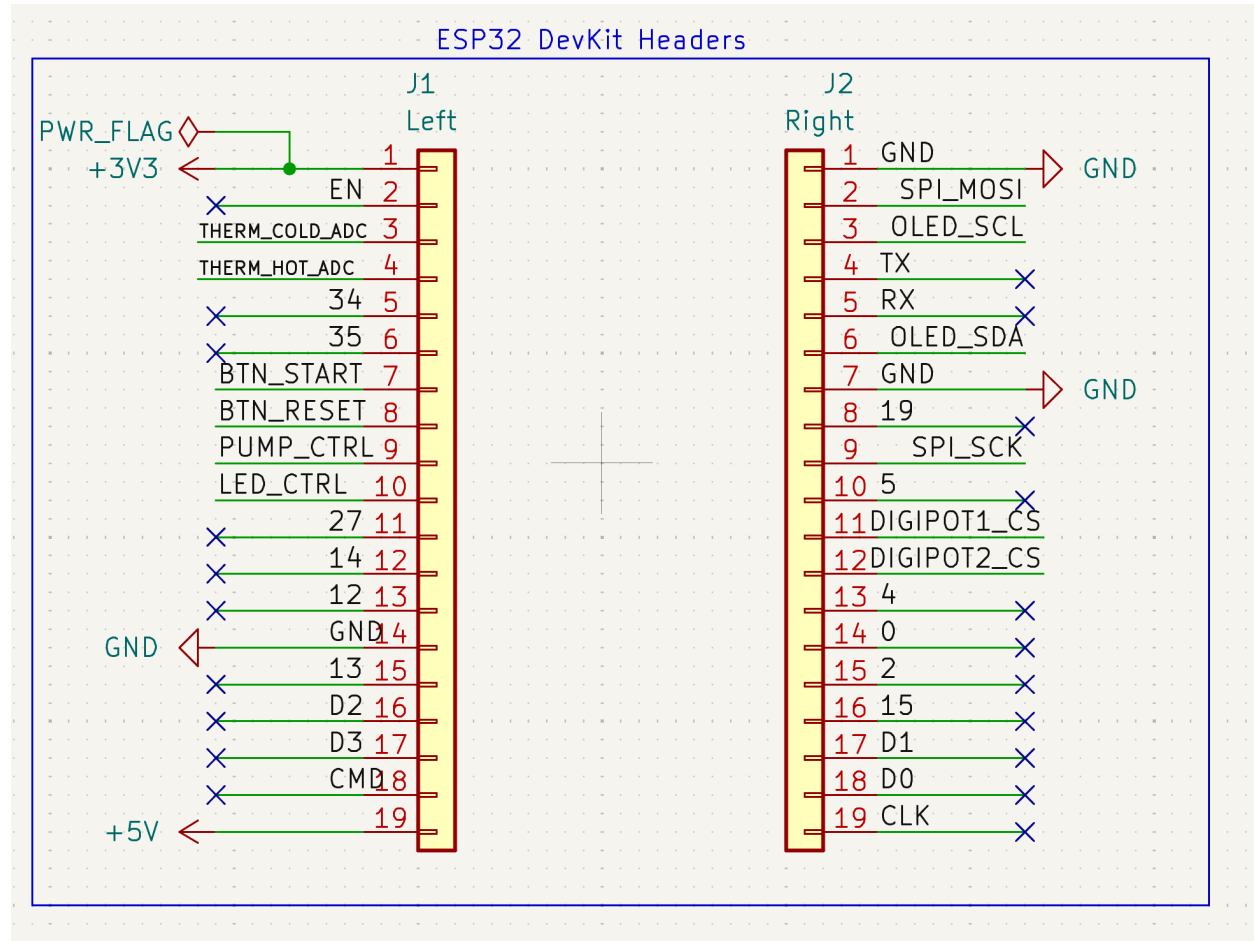


Figure 2. ESP32 DevKit headers.

The full 38-pin ESP32 DevKit was chosen to reuse its regulator, USB programming, reset and boot support circuits. The carrier uses two 1×19 socket headers; the exact board model, pinout and row spacing must match before layout.

The schematic uses two 1×19 headers:

J1 = ESP32 DevKit left header

J2 = ESP32 DevKit right header

The red numbers in KiCad are connector/header pin numbers. The functional labels correspond to the ESP32 DevKit pinout used during schematic development.

Used ESP32 pins

ESP32 DevKit pin label	Net name	Function in this PCB
3V3	+3V3	3.3 V logic rail from ESP32 DevKit
5V	+5V	5 V input to ESP32 DevKit from 12 V to 5 V buck converter
GND	GND	Common ground reference
VP / GPIO36	THERM_COLD_ADC	Analog ADC input for cold-side thermistor divider
VN / GPIO39	THERM_HOT_ADC	Analog ADC input for hot-side thermistor divider
GPIO32	BTN_START	Digital input for start button
GPIO33	BTN_RESET	Digital input for reset button
GPIO25	PUMP_CTRL	Digital output controlling alcohol pump MOSFET
GPIO26	LED_CTRL	Digital output controlling LED strip MOSFET
GPIO21	OLED_SDA	I2C data line for OLED
GPIO22	OLED_SCL	I2C clock line for OLED
GPIO23	SPI_MOSI	SPI data from ESP32 to digital potentiometers
GPIO18	SPI_SCK	SPI clock for digital potentiometers
GPIO17	DIGIPOT1_CS	Chip select for digital potentiometer 1
GPIO16	DIGIPOT2_CS	Chip select for digital potentiometer 2

Notes on communication pins

GPIO21 and GPIO22 were selected for I2C because they are commonly used as the default ESP32 I2C pins:

GPIO21 = SDA,

GPIO22 = SCL.

The ESP32 can route I2C to other GPIO pins in software, but using the common/default pins keeps the design conventional.

GPIO23 and GPIO18 were selected for SPI:

GPIO23 = MOSI, GPIO18 = SCK.

MISO was not included because the digital potentiometer concept only requires the ESP32 to send commands to the digital potentiometer ICs.

Each digital potentiometer gets its own chip-select line:

DIGIPOT1_CS, DIGIPOT2_CS.

This allows the same MOSI and SCK lines to be shared.

Unused ESP32 pins

Unused pins were marked with KiCad no-connect flags.

Important pins intentionally avoided:

D0, D1, D2, D3, CMD, CLK: these are typically related to flash memory on ESP32 modules and should not be used for normal project signals.

GPIO0, GPIO2, GPIO12, GPIO15: boot/strapping-sensitive pins, avoided for beginner-safe design.

TX and RX: not used because the DevKit handles programming/debug through USB.

EN: not used because the DevKit already has its own enable/reset circuitry.

Power Architecture

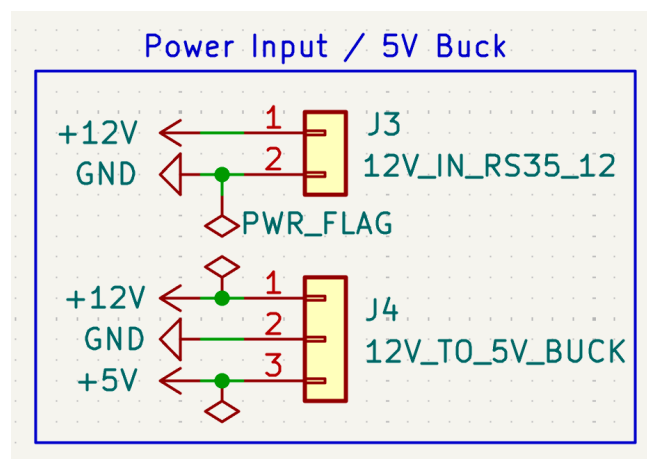


Figure 3. Power input and 5 V buck stage.

12 V input

Connector: J3 = 12V_IN_RS35_12.

Current schematic:

Pin	Net
1	+12V
2	GND

This represents the 12 V input from the Mean Well RS-35-12 supply.

The RS-35-12 12 V supply powers:

- 12 V to 5 V buck converter
- ESP32 DevKit indirectly through 5 V
- Alcohol pump
- LED strip
- Radiator fans
- Water pump.

Important design concern:

The 12 V supply power budget must be checked once all load currents are finalized. The water pump alone is close to 20 W, or about 1.67 A at 12 V. The alcohol pump is 5 W, or about 0.42 A. Fans, LED strip, and ESP32/buck current must also be included.

Total 12 V current estimate equation:

$$I_{\text{total}} = I_{\text{water_pump}} + I_{\text{alcohol_pump}} + I_{\text{fans}} + I_{\text{LED_strip}} + I_{\text{5V_buck_input}}$$

Known calculations:

$$\text{Alcohol pump: } I = P / V = 5 \text{ W} / 12 \text{ V} \approx 0.42 \text{ A}$$

$$\text{Water pump: } I = P / V = 20 \text{ W} / 12 \text{ V} \approx 1.67 \text{ A.}$$

If the 12 V supply is 3 A, the water pump plus alcohol pump already uses about 2.08 A, leaving limited margin for fans, LED strip, and electronics. This should be checked before final layout.

12 V to 5 V buck converter

Connector/block: J4 = 12V_TO_5V_BUCK.

Current schematic:

Pin	Net
1	+12V
2	GND
3	+5V

This represents a buck converter stage that takes 12 V and produces 5 V for the ESP32 DevKit.

Unresolved decision:

The buck converter implementation remains to be selected:

a module mounted directly on the PCB or an external buck module wired to the PCB or a fully designed buck converter circuit on the PCB.

This matters because the exact footprint and layout depend heavily on the selected option.

3.3 V rail

The ESP32 DevKit provides the +3V3 rail through its onboard regulator.

The PCB uses +3V3 for:

- thermistor dividers
- OLED module logic/power
- digital potentiometer logic, assuming compatible ICs
- possibly other low-voltage logic.

Important constraint:

ESP32 GPIO and ADC pins are 3.3 V logic. Do not apply 5 V or 12 V directly to GPIO/ADC pins.

Thermistor Subsystem

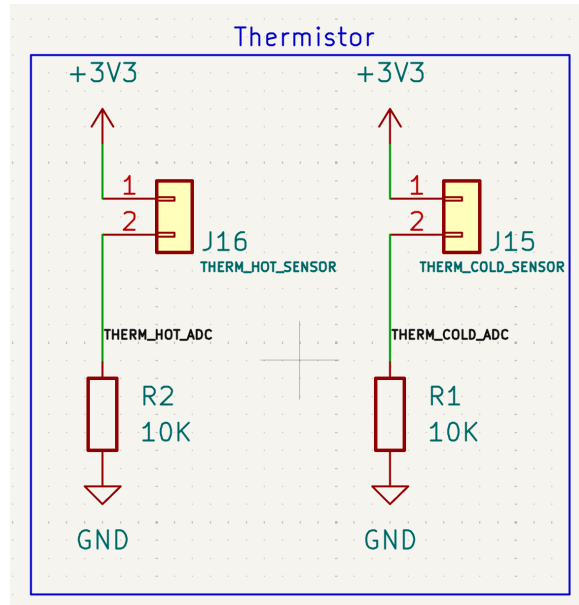


Figure 4. Thermistor sensor connectors.

Current revised design

After Nathaniel confirmed the thermistors will be physically mounted on the system, the thermistor symbols were removed and replaced with connectors.

Current connectors:

J15 = THERM_COLD_SENSOR, J16 = THERM_HOT_SENSOR.

Current cold thermistor circuit:

Component/net	Connection
J15 pin 1	+3V3
J15 pin 2	THERM_COLD_ADC
R1	10 kΩ from THERM_COLD_ADC to GND
ESP32 pin	VP / GPIO36 reads THERM_COLD_ADC

Current hot thermistor circuit:

Component/net	Connection
J16 pin 1	+3V3
J16 pin 2	THERM_HOT_ADC
R2	10 kΩ from THERM_HOT_ADC to GND
ESP32 pin	VN / GPIO39 reads THERM_HOT_ADC

Rationale

The ESP32 cannot directly read resistance. It reads voltage on ADC pins.

A thermistor changes resistance with temperature. With the voltage divider, the temperature-dependent resistance becomes a temperature-dependent voltage.

For the current topology:

+3.3 V → external NTC thermistor → ADC node → 10 kΩ resistor → GND

Voltage divider equation:

$$V_{\text{ADC}} = 3.3 \text{ V} \times R_{\text{fixed}} / (R_{\text{NTC}} + R_{\text{fixed}})$$

With $R_{\text{NTC}} = 10 \text{ k}\Omega$ and $R_{\text{fixed}} = 10 \text{ k}\Omega$:

$$V_{\text{ADC}} = 3.3 \text{ V} \times 10\text{k} / (10\text{k} + 10\text{k}) = 1.65 \text{ V}$$

For a 10 kΩ NTC thermistor:

temperature increases → thermistor resistance decreases → ADC voltage increases

temperature decreases → thermistor resistance increases → ADC voltage decreases.

The ESP32 ADC reads the voltage. Firmware then converts:

ADC voltage → thermistor resistance → temperature

The thermistor conversion requires the thermistor's resistance curve, beta value, or lookup table.

The exact thermistor beta value is still unresolved.

Alcohol Pump MOSFET Switch

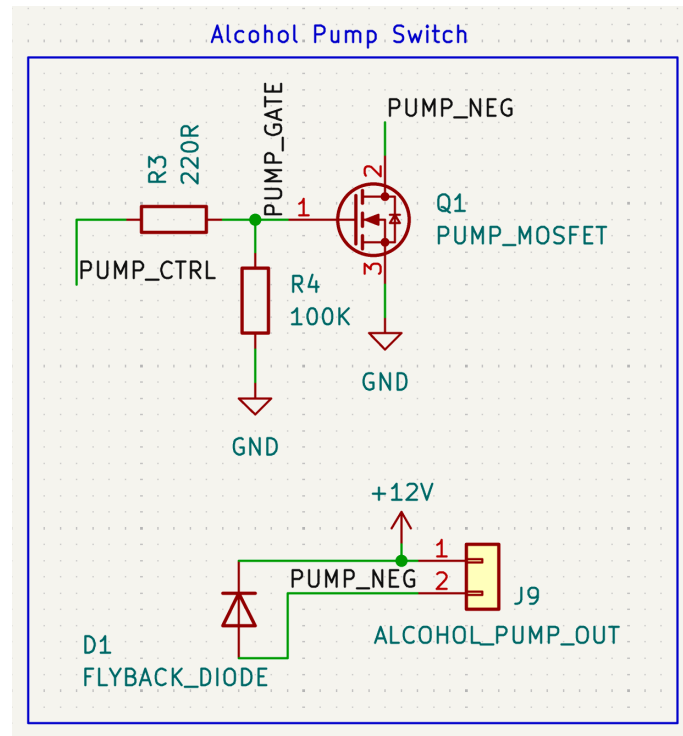


Figure 5. Alcohol pump MOSFET switch.

The alcohol pump requires on and off control only. No direction reversal is required, so an H-bridge is unnecessary.

Purpose

The alcohol pump needs to be controlled by the ESP32. The ESP32 GPIO cannot power the pump directly, so a MOSFET switch is used.

The alcohol pump is confirmed as approximately:

12 V, 5 W, current ≈ 0.42 A.

Calculation:

$$I = P / V = 5 \text{ W} / 12 \text{ V} \approx 0.42 \text{ A}$$

Current schematic

Main components:

Q1 = PUMP_MOSFET

R3 = 220 Ω and R4 = 100 k Ω

D1 = FLYBACK_DIODE

J9 = ALCOHOL_PUMP_OUT.

Connector:

J9 pin	Net
1	+12V
2	PUMP_NEG

MOSFET connections:

MOSFET terminal	Connection
Gate	PUMP_GATE
Drain	PUMP_NEG
Source	GND

Control signal:

ESP32 GPIO25 → PUMP_CTRL → R3 220 Ω → PUMP_GATE

Pulldown:

PUMP_GATE → R4 100 k Ω → GND

Flyback diode:

Diode side	Connection
Cathode/bar side	+12V
Anode side	PUMP_NEG

Operating principle

This is a low-side N-channel MOSFET switch.

“Low-side” means the MOSFET is placed between the load and ground:

+12V → pump → MOSFET → GND

When PUMP_CTRL is LOW:

gate voltage $\approx 0\text{ V}$ → MOSFET is off → drain-source path is open, so pump negative side is not connected to ground and pump is off.

When PUMP_CTRL is HIGH:

ESP32 drives gate to approximately 3.3 V → MOSFET turns on and drain-source path conducts so pump negative side connects to ground and current flows through the pump → pump turns on.

Gate resistor

$R3 = 220\ \Omega$

Purpose:

- limits the sudden current spike into the MOSFET gate
- reduces ringing/noise
- limits transient gate-charging current at the ESP32 GPIO
- makes switching cleaner.

The MOSFET gate behaves somewhat like a small capacitor, so the gate resistor slows the sudden charge/discharge slightly.

Pulldown resistor

$R4 = 100\text{ k}\Omega$

Purpose:

- keeps the MOSFET gate at 0 V when ESP32 is booting, resetting, or not actively driving the pin
- provides the intended default-off bias.

The pulldown is called “weak” because $100\text{ k}\Omega$ is a large resistance. It gently pulls the gate to ground, but when the ESP32 drives the gate high, the ESP32 easily overrides it.

Default state:

gate = 0 V → MOSFET off

Flyback diode

The alcohol pump is a motor/inductive load. When switched off, the motor current cannot stop instantly and can create a voltage spike at PUMP_NEG.

The flyback diode gives the leftover motor current a safe loop path.

Normal operation:

+12V side is high and PUMP_NEG side is near ground when MOSFET is on, so the diode is reverse-biased/off.

When the MOSFET turns off:

pump tries to keep current flowing and the PUMP_NEG can spike upward → diode turns on once PUMP_NEG rises above the +12 V side by about a diode drop → current safely circulates through pump and diode → the diode limits the turn-off transient.

The selected diode and MOSFET ratings still require verification.

Important correction made during discussion:

In this low-side switch, the dangerous spike at PUMP_NEG is usually a positive spike, not a negative spike.

LED Strip MOSFET Switch

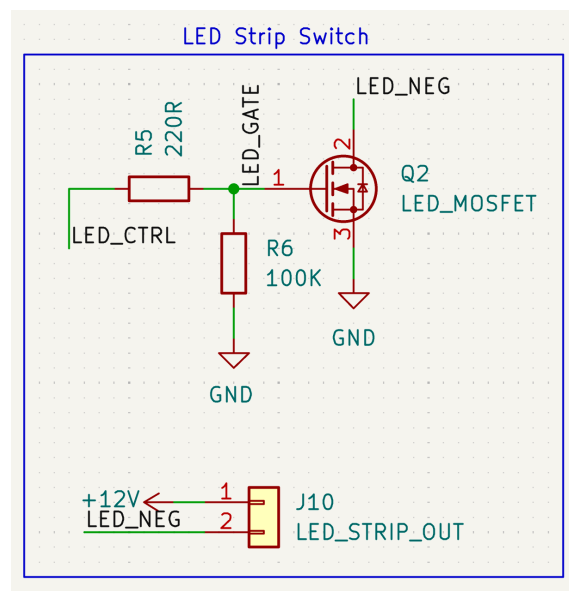


Figure 6. LED strip MOSFET switch.

Purpose

The LED strip is controlled by the ESP32 using a MOSFET switch.

The LED strip is intended to illuminate the chamber, so particle tracks are visible.

Current schematic

Main components:

Q2 = LED_MOSFET, R5 = 220 Ω , R6 = 100 k Ω , J10 = LED_STRIP_OUT.

Connector:

J10 pin	Net
1	+12V
2	LED_NEG

MOSFET connections:

MOSFET terminal	Connection
Gate	LED_GATE
Drain	LED_NEG
Source	GND

Control signal:

ESP32 GPIO26 $\rightarrow$ LED_CTRL $\rightarrow$ R5 220 Ω $\rightarrow$ LED_GATE

Pulldown:

LED_GATE $\rightarrow$ R6 100 k Ω $\rightarrow$ GND

Operating principle

Same as the alcohol pump switch, but without the flyback diode.

When LED_CTRL is LOW:

MOSFET off $\rightarrow$ LED negative side not connected to ground $\rightarrow$ LED strip off.

When LED_CTRL is HIGH:

MOSFET gate $\approx 3.3 \rightarrow$ MOSFET on $\rightarrow$ LED negative side connects to ground $\rightarrow$ current flows: +12V $\rightarrow$ LED strip $\rightarrow$ MOSFET $\rightarrow$ GND $\rightarrow$ LED strip on.

Why no flyback diode

The LED strip is not a motor/inductive load, so it does not require a flyback diode like the pump.

MOSFET Selection Requirements

The schematic currently uses generic N-channel MOSFET symbols for Q1 and Q2.

These are not final part numbers.

The final MOSFETs must be:

N-channel, logic-level, fully enhanced with 3.3 V gate drive, rated for the load current with margin, rated for at least the relevant voltage, preferably 30 V or more for 12 V loads and low RDS(on) at VGS = 2.5 V or VGS = 3.3 V.

Important clarification:

VGS (th) is not enough. Gate threshold voltage only means the MOSFET barely starts to turn on. It does not mean the MOSFET can carry the load current efficiently.

Need to check:

RDS (on) specified at low gate voltage.

RDS (on) means drain-source resistance when the MOSFET is on. Lower RDS (on) means less heat.

MOSFET heat estimate:

$$P_{\text{MOSFET}} \approx I^2 \times RDS(\text{on})$$

This will be useful when choosing the final MOSFETs after load currents are known.

Water Pump Output

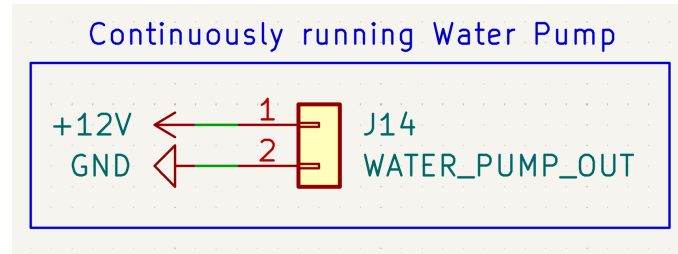


Figure 7. Continuous water pump output.

Purpose

The water pump circulates coolant/water in the hot-side cooling loop. It is not for putting water into the chamber.

The water loop carries heat away from the hot side of the Peltier modules to the radiator.

Current design

Current decision: water pump runs continuously.

Connector:

J14 = WATER_PUMP_OUT.

Current schematic:

J14 pin	Net
1	+12V
2	GND

No MOSFET control is used for the water pump.

Power and current

Nathaniel stated the water pump draws closer to 20 W.

At 12 V: $I = 20 \text{ W} / 12 \text{ V} \approx 1.67 \text{ A}$

Based on Nathaniel's rule, this should use a screw terminal because it is above 1 A.

Radiator Fan Outputs

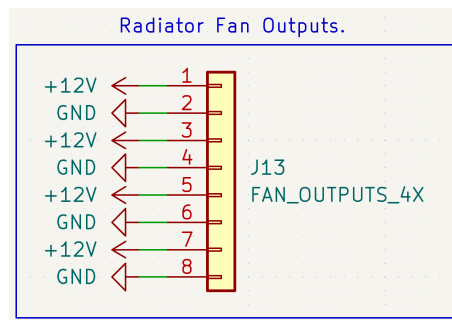


Figure 8. Radiator fan outputs.

Purpose

Fans cool the radiator. The water loop carries heat from the Peltier hot side to the radiator, and fans blow air through the radiator to dump that heat into room air.

Cooling chain:

Heat passes from the cold plate through the Peltier stack into the water block. The water pump circulates coolant to the radiator, where the fans reject heat to ambient air.

Current schematic

Connector:

J13 = FAN_OUTPUTS_4X.

Current schematic supports four 12 V fan outputs:

Fan	Pins
Fan 1	pin 1 = +12V, pin 2 = GND
Fan 2	pin 3 = +12V, pin 4 = GND
Fan 3	pin 5 = +12V, pin 6 = GND
Fan 4	pin 7 = +12V, pin 8 = GND

Current decision: fans run continuously from 12 V.

No MOSFET control is included for fans.

OLED Display Subsystem

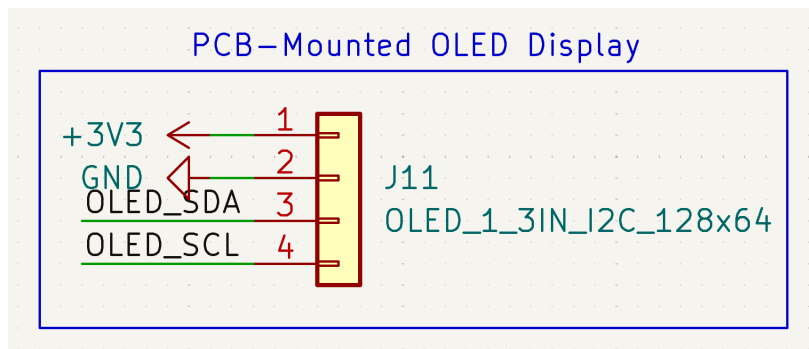


Figure 9. PCB mounted OLED interface.

Revised design

Nathaniel said the PCB is mounted on one of the side panels, so the OLED and buttons can be directly on the PCB. The OLED design was updated to be PCB-mounted.

Current connector/module interface:

J11 = OLED_1_3IN_I2C_128x64.

Current schematic:

J11 pin	Net
1	+3V3
2	GND
3	OLED_SDA
4	OLED_SCL

The schematic block was renamed:

PCB-Mounted OLED Display.

OLED selection direction

A 2-inch OLED was considered, but concern was raised that larger OLEDs are more likely to use SPI or have more complex pinouts. To minimize schematic changes and keep the design simple, the direction was changed back to:

1.3-inch; 128×64 pixels; I2C interface; 4-pin module; 3.3 V-compatible.

Reason:

1.3-inch 128×64 I2C OLED modules are common and fit the current simple 4-wire design.

Important OLED warning

The electrical schematic can remain as 4 pins, but the exact OLED module must be selected before layout because modules may differ in:

pin order, module board size, mounting hole spacing, header orientation, voltage requirement, I2C vs SPI interface etc.

Some modules may order pins as VCC, GND, SCL, SDA instead of VCC, GND, SDA, SCL. The final footprint must match the actual module.

Assembly note

PCB-mounted does not automatically mean the manufacturer will solder the OLED module during PCB fabrication.

There are two cases:

Bare PCB order: OLED is soldered/mounted manually later

PCBA order: manufacturer assembles it only if the part/module is included and supported.

This decision is still not finalized.

Button Subsystem

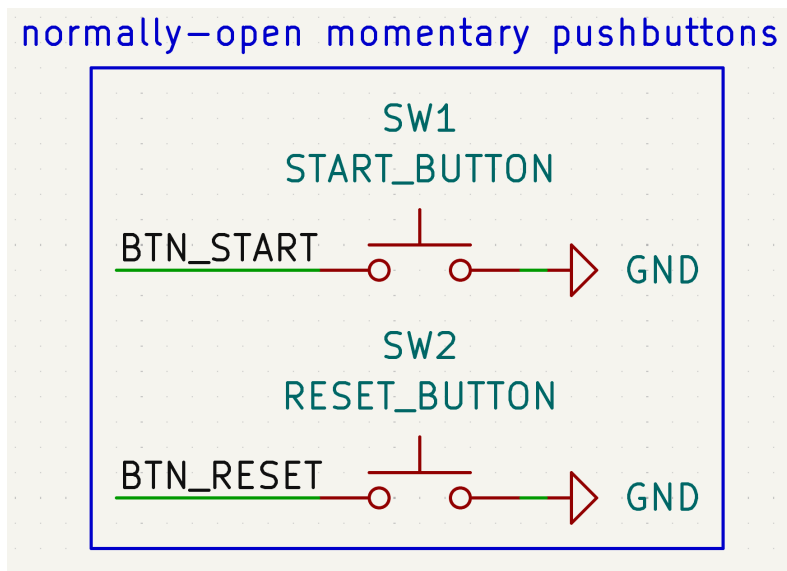


Figure 10. Start and reset buttons.

BTN_RESET is an application input on GPIO33, not a hardware connection to the ESP32 EN pin. Its reset action must be defined in firmware.

Revised design

After Nathaniel confirmed OLED/buttons can be PCB-mounted, J12 was replaced with two real switch symbols.

Current design:

SW1 = START_BUTTON and SW2 = RESET_BUTTON.

Both are normally-open momentary pushbuttons.

Current wiring:

Button	Connection
SW1 START_BUTTON	BTN_START to GND
SW2 RESET_BUTTON	BTN_RESET to GND

Operating principle

The ESP32 will use internal pull-up resistors in firmware.

Button not pressed:

switch open → ESP32 pin is pulled HIGH internally → code reads HIGH.

Button pressed:

switch closes → signal connects to ground → ESP32 reads LOW.

This is an active-low button configuration.

Digital Potentiometer PSM-292 Control Subsystem

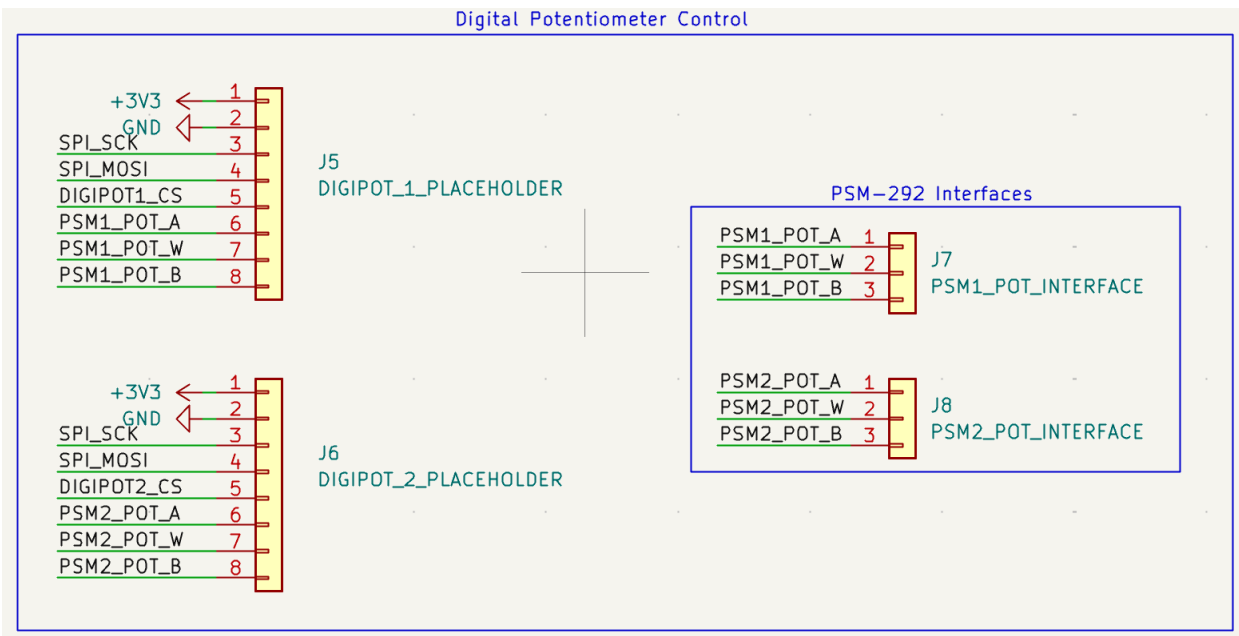


Figure 11. Digital potentiometer and PSM 292 interface.

Compatibility with the PSM-292 adjustment circuit is the main unresolved schematic dependency.

Purpose

The project uses two PSM-292 adjustable buck converter modules. These control the output voltage going to Peltier stages/groups.

The PSM-292 modules normally use manual potentiometers to adjust output voltage.

The design idea is to electronically replace or interface with those manual potentiometers using digital potentiometer ICs controlled by the ESP32.

Important correction

The PSM-292 itself does not understand SPI.

The digital potentiometer IC understands SPI.

Correct control chain:

ESP32 → SPI → digital potentiometer IC → A/W/B terminals → PSM-292 potentiometer pads → converter output voltage → Peltier cooling power.

Current placeholder design

Current schematic placeholders:

J5 = DIGIPOT_1_PLACEHOLDER and J6 = DIGIPOT_2_PLACEHOLDER.

Each has 8 pins:

Pin	Function
1	+3V3
2	GND
3	SPI_SCK
4	SPI_MOSI
5	DIGIPOTx_CS
6	PSMx_POT_A
7	PSMx_POT_W
8	PSMx_POT_B

Where:

J5 uses DIGIPOT1_CS, PSM1_POT_A/W/B and J6 uses DIGIPOT2_CS, PSM2_POT_A/W/B.

Proposed next schematic revision

It was later recognized that J5 and J6 should probably not remain as external 8-pin connectors.

Proposed direction, pending IC selection and compatibility testing:

Replace J5 with actual digital potentiometer IC symbol, likely U2 and replace J6 with actual digital potentiometer IC symbol, likely U3.

Then:

SPI lines remain internal PCB traces; A/W/B outputs go to external PSM-292 interface connectors.

PSM-292 interface connectors

Current connectors:

J7 = PSM1_POT_INTERFACE and J8 = PSM2_POT_INTERFACE.

Each has three pins:

Pin	Net
1	PSMx_POT_A
2	PSMx_POT_W
3	PSMx_POT_B

These are intended to connect from the control PCB to the physical PSM-292 module's potentiometer pads.

These are signal/control connections, not high-current Peltier power connections.

Why two digital potentiometers

There are two PSM-292 modules. Each module needs its own electronic potentiometer control.

Therefore:

digital potentiometer 1 controls PSM-292 #1 and digital potentiometer 2 controls PSM-292 #2.

The SPI clock and MOSI lines are shared. Each digital potentiometer has a separate chip-select pin.

Critical unresolved technical checks

Before finalizing the digital potentiometer ICs, the following must be confirmed or measured:

- Resistance value of the original PSM-292 manual potentiometer
- Voltage range on the PSM-292 potentiometer terminals A/W/B
- Current through the potentiometer terminals
- Whether the manual potentiometer can be safely removed/desoldered
- Whether a digital potentiometer can directly replace it
- Whether one PSM-292 should be tested first before final PCB design
- Exact digital potentiometer IC part number
- Whether the digital potentiometer terminals can tolerate the PSM-292 voltage range.

Important safety/design constraint:

Many digital potentiometers powered at 3.3 V can only tolerate terminal voltages between GND and VDD. If the PSM-292 potentiometer terminals exceed 0–3.3 V, a normal 3.3 V digital potentiometer could be damaged.

Considered IC family

A Microchip MCP41xxx/MCP42xxx-style SPI digital potentiometer family was mentioned as a possible direction, but no IC was finalized.

Do not treat any digital potentiometer IC as selected yet.

Connector Strategy

Nathaniel provided a general connector rule:

loads under 1 A → pin headers acceptable

loads higher than 1 A → screw terminals.

Current connector implications

Interface	Current expectation
ESP32 DevKit headers	1×19 female headers, exact spacing required
Alcohol pump output	likely pin header, because 0.42 A
Water pump output	screw terminal, because about 1.67 A

Interface	Current expectation
Fan outputs	likely pin headers if each fan is under 1 A; exact fan current still needed
LED strip output	depends on final LED strip current
Thermistor connectors	small pin headers or JST-style connectors; tiny current
PSM-292 A/W/B interface	small pin header or JST-style connector; signal-level wiring
12 V input J3	likely screw terminal because it carries total 12 V system current
OLED module	module header/footprint matching selected OLED
Pushbuttons	actual PCB-mounted button footprints, not connector
Buck converter J4	depends on whether module is onboard, external, or fully designed on PCB

Pin header clarification

When Nathaniel says, “pin headers,” he most likely means common 2.54 mm header pins, usually male pins soldered to the PCB with a female cable connector on the wire side.

However, pin header can technically mean either male or female. The exact choice must be made during footprint assignment.

For the ESP32 DevKit, the PCB likely needs female socket headers, so the DevKit plugs into the PCB.

Electrical rule check notes

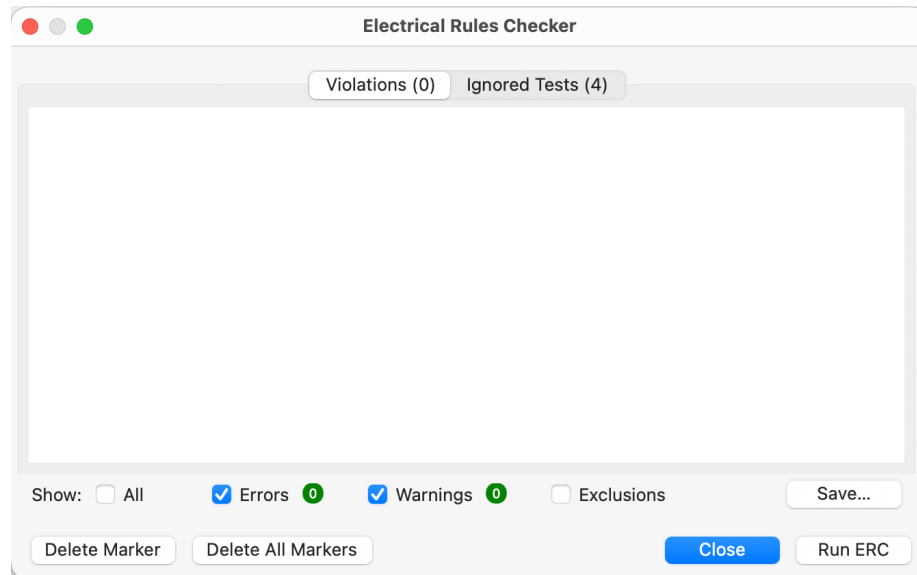


Figure 12. Electrical rule check result.

ERC was run during schematic cleanup.

Issues encountered and resolved:

Duplicate net names

Example:

+5V power symbol and 5V net label on the same wire.

KiCad warning:

both +5V and 5V attached to the same item.

Fix:

remove extra 5V net label and keep +5V power symbol.

Similar cleanup applied to extra labels like VP and VN when they were placed as net labels on the same wires as THERM_COLD_ADC and THERM_HOT_ADC.

Important distinction:

pin names built into the symbol can remain

duplicate net labels placed on wires should be removed.

Unconnected wire endpoints

Some warnings were caused by wires or labels appearing visually connected but not actually snapped to the pin/wire grid.

Fix:

move symbols or labels so wires snap properly, re-run ERC.

Power input not driven

KiCad complained:

“Input Power pin not driven by any Output Power pins”

Reason:

KiCad saw power nets like +12V, +5V, +3V3, and GND, but generic connectors do not automatically tell KiCad that a real supply is driving those nets.

Fix:

add PWR_FLAG symbols on the real power rails: +12V; +5V; +3V3; GND.

No-connect flags

No-connect flags were placed on unused ESP32 pins.

Important rule established:

Do not place no-connect flags on pins that are used through net labels. For used signal pins like SPI_MOSI, OLED_SCL, OLED_SDA, SPI_SCK, DIGIPOT1_CS, and DIGIPOT2_CS, the correct fix is proper net-label connection, not no-connect flags.

Reported ERC result

After fixes, ERC showed:

0 violations; 0 errors; 0 warnings.

Ignored tests were present. Review their scope when rerunning ERC, a clean ERC result does not establish component compatibility or hardware performance.

Design decision record

Confirmed design direction

- Use ESP32 DevKit carrier-board approach, not bare ESP32 chip.
- Use 38-pin ESP32 DevKit layout with two 1×19 headers.
- Use ESP32 ADC pins VP/GPIO36 and VN/GPIO39 for thermistor voltages.
- Use GPIO25 for alcohol pump MOSFET control.
- Use GPIO26 for LED strip MOSFET control.
- Use GPIO21/GPIO22 for I2C OLED.
- Use GPIO23/GPIO18 for SPI MOSI/SCK.
- Use GPIO17/GPIO16 as chip selects for the two digital potentiometers.
- Alcohol pump controlled by low-side N-channel MOSFET.
- LED strip controlled by low-side N-channel MOSFET.
- Water pump runs continuously from 12 V.
- Radiator fans run continuously from 12 V.
- Thermistors are mounted physically on the system, not directly on the PCB.
- Fixed 10 kΩ thermistor divider resistors remain on PCB.
- Buttons are PCB-mounted normally-open momentary pushbuttons.
- OLED is PCB-mounted, likely 1.3-inch 128×64 I2C module.
- Loads under 1 A can use pin headers; loads above 1 A should use screw terminals.

Considered but rejected or changed

- Bare ESP32 chip/module: rejected in favor of full ESP32 DevKit.
- External OLED/buttons via connector only: changed after Nathaniel said OLED/buttons can be on PCB side panel.
- Thermistors directly on PCB: changed because thermistors must measure actual physical hot/cold locations.
- 2-inch OLED: considered but moved away from because larger OLED modules may use SPI or more complex pinouts. 1.3-inch I2C retained as the working target; the exact module is still unconfirmed.
- J5/J6 as external digital potentiometer connectors: recognized as probably not final. They should likely become actual digital potentiometer ICs on the PCB.
- Water pump MOSFET control: not included for now; water pump is continuous.

- Fan MOSFET control: not included for now; fans are continuous.

Open items before release

Resolve these items before committing the affected footprints, routing or manufacturing release.

ESP32 DevKit

Confirm the exact 38-pin DevKit model, electrical pinout and mechanical dimensions.

Required details:

row-to-row spacing; pin pitch; USB port location and clearance; board outline etc.

OLED

Need exact OLED module selection.

Current target:

1.3-inch; 128×64; I2C; 3.3 V-compatible; PCB-mounted.

Need confirm:

exact module dimensions; pin order; footprint/header spacing; mounting holes; side-panel cutout size.

LED strip

Still TBD.

Need:

voltage; current or wattage; physical connector/wire style; whether it is 12 V; length/brightness requirement; whether it should be switched only on/off or dimmed with PWM.

Radiator fans

Need final fan part number and quantity.

Current schematic supports 4 fan outputs. BOM screenshots have shown inconsistent/unclear fan quantity/part status.

Need:

exact 120 mm fan part; voltage; current per fan; connector style; final fan quantity.

Digital potentiometer and PSM 292

Major unresolved item.

Need:

original PSM-292 potentiometer resistance value; terminal voltages on A/W/B; terminal current; confirmation that manual pot can be removed/desoldered; test result showing digital pot replacement works; exact digital potentiometer IC; whether digital pot terminal voltage range is safe; footprint/package for the digital potentiometer IC, whether any extra buffering/protection/isolation is needed.

Buck converter

Need exact decision:

onboard module; external module; or full onboard buck circuit.

Need exact part if module-based.

The footprint for J4 cannot be finalized until this is known.

12 V power budget

Need final current values for:

water pump; alcohol pump; fans; LED strip; buck converter input current; ESP32/OLED/digital pot consumption.

Then verify RS-35-12 supply can handle total load with margin.

Final parts still to choose

Not yet finalized:

logic-level N-channel MOSFETs for pump and LED; flyback diode for alcohol pump; exact connector footprints; pushbutton footprints; resistor package sizes; thermistor model, connector style; digital potentiometer ICs; OLED module footprint; ESP32 DevKit header footprints and spacing; buck converter footprint/implementation.

Next steps and PCB layout

The project is still pre-layout. Remaining work:

Finalize schematic changes

Still required:

replace J5/J6 digital potentiometer placeholders with real IC symbols once selected; keep J7/J8 as PSM-292 interface connectors; update LED strip once final part/rating is known; update fan outputs if final fan count differs from current 4-output design; update buck converter block once exact implementation is chosen.

Assign footprints

Need footprint assignment for:

ESP32 DevKit 1×19 female headers; OLED module/header; SW1/SW2 pushbuttons; J3 12 V input connector; J4 buck converter/module interface; J7/J8 PSM-292 potentiometer interfaces; J9 alcohol pump output; J10 LED strip output; J13 fan outputs; J14 water pump output; J15/J16 thermistor connectors; MOSFETs Q1/Q2; flyback diode D1; resistors R1/R2/R3/R4/R5/R6; PWR_FLAG symbols do not need physical footprints.

Board placement planning

Nathaniel suggested:

OLED and buttons on one side of PCB facing the user/side panel cutout; microcontroller, connectors, and chips on the other side of PCB.

Layout must account for:

side-panel cutouts; OLED visibility; button accessibility; ESP32 USB access; wire routing to pumps/fans/thermistors/PSM modules; connector accessibility; mechanical mounting holes; enclosure clearance.

Routing considerations

Important layout considerations:

- Use wider traces for 12 V load currents.
- Water pump trace must handle about 1.67 A plus margin.
- 12 V input trace may carry total system auxiliary current and should be sized accordingly.

- Keep MOSFET current paths short and wide.
- Place flyback diode close to the pump connector/MOSFET loop.
- Place gate resistors near MOSFET gates.
- Place pulldown resistors near MOSFET gates.
- Keep analog thermistor traces away from noisy high-current switching paths where possible.
- Use ground pours if appropriate.
- Ensure common ground between ESP32 control board and relevant control electronics.
- Do not route high-current Peltier power through this PCB.
- Run DRC after routing.
- Check 3D viewer for connector orientation, side placement, OLED position, and ESP32 fit.

Manufacturing outputs still needed

After layout:

DRC clean; silkscreen labels checked; connector pin labels visible where useful; Gerber files; drill files; BOM; pick-and-place file if PCBA is used; assembly notes; 3D view screenshot for review; final review with project lead before ordering.

Design calculations and verification plan

The existing numerical work comprises the thermistor divider midpoint and nominal pump-current estimates. These support early component and power planning; trace widths, MOSFET dissipation and full supply loading are not yet finalized.

Calculation	Current value	Meaning
Alcohol pump	$5\text{ W} / 12\text{ V} \approx 0.42\text{ A}$	Nominal estimate: startup current remains to be checked.
Water pump	$20\text{ W} / 12\text{ V} \approx 1.67\text{ A}$	Approximate load; rating needs confirmation.
Combined pumps	$25\text{ W} / 12\text{ V} \approx 2.08\text{ A}$	Calculated from unrounded inputs. The earlier 2.09 A figure sums rounded values.
Divider midpoint	$3.3\text{ V} \times 10 / (10 + 10) = 1.65\text{ V}$	Equal 10 k Ω thermistor and fixed resistor.
MOSFET conduction loss	$P \approx I^2 \times R_{DS(on)}$	No final result until parts and load currents are known.

The 10 k Ω divider is the current schematic choice. Equal resistance gives a midpoint output. Sensitivity and useful ADC range at the actual hot and cold operating temperatures still need validation with the selected thermistor curve. The handoff's conditional 3 A supply example is a planning assumption, not a verified supply rating.

Planned hardware validation

The electronics requirements call for collecting parts and breadboard testing before PCB manufacture. Record the PSM-292 potentiometer measurements and replacement test first. Then verify regulated rails, thermistor readings, active-low buttons, OLED operation and pump and LED switching. Add actual load currents, startup behavior and measured temperatures as results become available. No completed bench-validation result is recorded here.

Maintaining the design record

Maintain this report alongside the PCB. For each resolved item, record the selected part, rationale, schematic revision and supporting measurement. Preserve earlier decisions as history. Add routed PCB views, assembly photographs and test results when available. Date each screenshot and test result so CNES can identify the design revision it represents.

Revision	Date	Change
0.2	12 September 2026	Clearer development status and document maintenance guidance.
[Next]	[Date]	[Record schematic, layout or test changes]